Physical
1/21/11

In molecular electronic spectroscopy:

1.) Regarding the Franck-Condon principle:

a) What is the Franck-Condon principle? (Use the transition from the electronic ground state of a diatomic molecule to an excited electronic state as an example to answer this question and include a simple graph in your answer.)

b) Derive a formula of the *Franck-Condon factor* from the *transition dipole moment*, and describe what the Franck-Condon factor states about the observed intensities of simultaneous electronic and vibrational transitions.

2.) Describe the sequence of steps leading to fluorescence.

3.) Describe the sequence of steps leading to phosphorescence.

4.) Describe how electronic excitation can lead to the phenomena of a) *molecular dissociation* and b) *predissociation*. How is predissociation detected in a spectrum?

5.) Compare the absorption spectrum of a transition between two molecular electronic states with the associated fluorescence spectrum which involves the same electronic states:

a) In which way is the information contained in the vibrational structure of an absorption spectrum different from the information contained in the vibrational structure of the associated fluorescence spectrum? (Draw a simple, qualitative graph showing wavelengths and intensities of a typical absorption spectrum and the associated fluorescence spectrum.)

b) In which way does an absorption spectrum differ from its associated fluorescence spectrum, and where are the two coincident?

Credits: no.1, 40 points; nos. 2, 3, 4a, 4b, 5a and 5b, 10 points each.